



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE
NAME

ANSWER SCHEME

CIVICS
GROUP

1 7 S

INDEX
NUMBER

H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

17 September 2018

2 hours

Candidates answer on the Question Paper.

No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **ONE** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use

Section A

1 / 29

2 / 13

3 / 8

Section B

4 or 5 / 25

Total / 75

Section A

Answer **all** the questions in this section.

QUESTION 1

DNA analysis revealed that 47 young children buried more than 1500 years ago near Rome in Italy died during an epidemic of an infectious disease. The nucleotide sequence of the 18S rRNA gene of *Plasmodium falciparum* was positively identified in some of the bones. *P. falciparum* is a parasite that causes the disease malaria.

- (a) (i) Describe the process by which repeated cycles of PCR could specifically amplify the *P. falciparum* 18S rRNA gene nucleotide sequences from a small quantity of the DNA from the buried bones. [4]

- [Compulsory] 2 primers complementary to the flanking region of each strand of the 18S rRNA gene sequences are used for PCR.

During PCR,

- Temperature is raised to 95°C (A: 92 – 98°C), where double stranded DNA denatures / hydrogen bonds between complementary base pairs are broken to produce two single strands
- Temperature is cooled to 55°C (A: 50 - 60°C), where the primers binds to the single stranded DNA by complementary base pairing. [mark once]
- Temperature raised to 72°C, where Taq polymerase elongates the primers by adding deoxyribonucleotides to the 3'OH end of the new complementary strand.
- .

- (ii) Suggest why the 18S rRNA gene sequence is used for the positive identification of *P. falciparum*. [2]

- rRNA gene codes for rRNA, which is required for the assembly of functional ribosome.
- Hence, DNA sequence of 18S RNA gene will be highly conserved (similar to the parasites found today).

P. falciparum infects a person when a mosquito carrying the parasite injects an anti-coagulant into the bloodstream prior to feeding on blood. The parasites multiply in liver cells before emerging after 9-30 days wrapped in liver cell surface membrane. They enter red blood cells, multiply and then cause rupture of the host cells and release of more parasites every 36-48 hours, in a manner that has some similarity to that of viruses.

- (b) (i) The symptoms of malaria occur repeatedly every 36-48 hours and include body pain. These symptoms are associated with many other infectious diseases, including those caused by the viruses.

Suggest how *P. falciparum* may cause these types of symptoms. [2]

[Why symptoms repeat every 36-48h]

- Due to repeated/periodic/recurrent release of parasites which trigger innate immune response e.g. release of cytokines, phagocytosis

[Why pain]

- Pain caused by damage to tissue / inflammation due to release of cytokines by immune cells / histamine release

- (ii) Use the information given and your own knowledge, suggest why it is extremely difficult to design an effective vaccine against malaria. [3]

- Parasite evades immune system by residing inside host liver cells for 9-30 days.
- [idea of] Uses liver cell membrane to disguise as 'self' to trick the immune system
- B cells / T helper cells generated from vaccination unable to use its B-cell receptor / T-cell receptor to bind to malarial antigens hence not stimulated to produce antibody and cell-mediated response respectively
- There is a large degree of antigenic variation in the parasite population due to different antigens expressed during each part of the parasite's life cycle.

- (c) Malaria was common in Italy, a European country situated in the Northern Hemisphere. Widespread land drainage together with the use of the insecticide DDT and the drug chloroquine eradicated both the mosquito vectors and the malarial parasites in the 1950s. Due to the success of these measures, they were later discontinued as they were no longer necessary.

Articles in the scientific literature more recently show that malarial mosquitoes are returning to Italy and increasing their numbers and their northerly range, with some cases of malaria being reported. In general, winters are milder and summers are hotter in the south of the country, with temperatures decreasing to the north, especially in the winters.

Discuss whether the return of malaria to Italy can be attributed to climate change. [4]

[Yes]

- Range extending further north may relate to warmer temperatures
- [idea of] higher temperature lead to increase in number of mosquitoes due to e.g. hasten the life cycle of mosquitoes due to increased metabolism
- Climate change leads to increase in rainfall may result in more flooded areas for mosquitoes to breed

[No]

- Return of malaria was due to the discontinued use of DDT
- [idea of] Mosquitos could be introduced into Italy from the surrounding countries
- [idea of] The parasite thrive and spend most of their life cycle in host body temperature (37°C) hence ambient temperature has hardly any effect on their survival

- (d) Chloroquine-resistant (CQR) malarial parasites, *P. falciparum*, were first reported in 1950s and are now widespread. The resistance is caused by mutations of a gene known as *pfcr*.

- (i) Explain whether resistance to chloroquine in *P. falciparum* is an example of continuous or discontinuous variation. [2]

- Discontinuous variation
- It involves only a single gene
- Different alleles have large effect/ gives discrete groups of resistant or non-resistant parasite

(ii) Explain why CQR *P. falciparum* are now widespread. [4]

- Variation in the population of *P. falciparum* due to random mutations
- Use of chloroquine acts as selection pressure on the *P. falciparum* population
- Chloroquine-resistant *P. falciparum* are at selective advantage and are likely to survive to reproduce
- ..and pass their CQR allele/ mutated *pfcr* allele to their fertile and viable offspring
- Allele frequency for CQR allele/ mutated *pfcr* increases over many generations within the *P. falciparum* population.

Analysis of *pfcr* in CQR *P. falciparum* from different parts of the world shows differences in one section of the gene only. The amino acid sequences coded for by this section are shown in Table 1.1, together with the amino acid sequence coded for by the allele found in parasites that are still susceptible to chloroquine (non-CQR). The amino acid sequence coded for by the rest of the gene is the same in all cases.

Table 1.1

allele of <i>pfcr</i> gene	amino acid sequence coded for by allele of <i>pfcr</i> gene
non-CQR	—Cys—Met—Asn—Lys—His—Ala—Glu—Asn—Ile —Met—
CQR from Africa	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Ile —Ile —
CQR from Asia	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Thr—Ile —
CQR from Peru and Brazil	—Ser—Met—Asn—Thr—His—Ser—Gln—Asp—Leu—Arg—
CQR from Colombia	—Cys—Met—Glu—Thr—Gln—Ser—Gln—Asn—Ile —Thr—

(iii) With reference to Table 1.1, calculate the mean number of different amino acids coded for by CQR alleles in comparison with the non-CQR allele. Show your working. [2]

$$\frac{6 + 7 + 7 + 6}{4} \quad [1]$$

$$= 6.5 \quad [1]$$

mean number of different amino acids

In the non-CQR *P. falciparum*, chloroquine accumulates in the digestive vacuole of *P. falciparum* and interferes with the detoxification of haem in the host red blood cell. This results in parasite death.

Research shows that there is a point mutation in the *pfcr* gene of CQR *P. falciparum* which leads to the formation of a chloroquine-resistance transporter located in the digestive vacuole membrane of CQR *P. falciparum*.

This mutation changes the amino acid from lysine to threonine at the binding site of the chloroquine-resistance transporter.

(e) Using the information provided, suggest how this mutation of *pfcr* gene increases the chloroquine resistance in *P. falciparum*. [3]

- Base-pair substitution of a nucleotide at the 1st or 2nd nucleotide of the triplet in the *pfcr* gene of CQR *P. falciparum*
- change in mRNA codon that codes for threonine instead of lysine
- ..alters the shape of the binding site of *pfcr* protein to become complementary to chloroquine / change the charge of the binding site in *pfcr* protein to allow binding to chloroquine
- ..chloroquine is transported out of the digestive vacuole.

In 2014, the World Health Organisation (WHO) estimated that 3200 million people were at risk of malaria. This was almost half of the world population in 2014.

Table 1.2 shows the number of cases of malaria and the number of deaths from malaria between 1998 and 2013. The table shows numbers for all countries of the world and for the countries in the WHO African region.

The table also shows the numbers in the African region as percentages of the numbers for all countries.

Table 1.2

year	number of cases of malaria in millions		cases in the African region as a percentage of all countries/ %	number of deaths from malaria in thousands		deaths in the African region as a percentage of all countries/ %
	all countries	African region		all countries	African region	
1998	272.9	237.6	87.1	1067.0	961.0	90.1
2003	236.0	186.6	79.1	872.0	800.0	91.7
2008	225.1	181.0	80.4	747.0	677.0	90.6
2013	198.0	158.4	80.0	584.0	525.6	90.0

(f) Describe the trends shown in Table 1.2.

[3]

- From 1998 to 2013, numbers of cases of malaria in millions have decreased in all countries from 272.9 to 198.0 and in Africa from 237.6 to 158.4.
- From 1998 to 2013, number of deaths from malaria in thousands have decreased steeply in all countries from 1067.0 to 584.0 and in Africa from 961.0 to 525.6.
- From 1998 to 2003, cases in Africa as a percentage of all countries decreases gradually from 87.1 to 79.1 and remains constant/ reaches a plateau of 80.0 from 2008 to 2013.
- From 1998 to 2013, deaths in Africa as a percentage of all countries fluctuates slightly between 90.0 and 91.7. A: remain constant

[Total: 29]

QUESTION 2

Some mothers of infants with Down syndrome have a mutation of a gene coding for an enzyme involved in the metabolism of methyl groups. A lack of methyl groups on the centromeric DNA results in mistakes in the segregation of chromosomes during cell division.

(a) Suggest how a lack of methyl groups on the centromeric DNA results in mistakes in segregation of chromosomes during cell division. [2]

- Decondensation/ uncoiling of the centromere region
- Shape of centromere is no longer complementary to kinetochore protein and thus cannot bind to centromere
- Spindle fibres cannot bind to the kinetochore for homologous chromosomes/ sister chromatids to separate properly during cell division

The gene mutation referred to in (a) also affects the metabolism of the vitamin folic acid. Abnormal folic acid metabolism is seen in mothers of infants with neural tube disorders (NTD).

Two groups of families, in Israel and Ukraine, were investigated to see whether there was any link between the incidence of Down syndrome and NTD. The results are shown in Table 2.1.

Table 2.1

group of families	number of pregnancies	number of cases of Down syndrome	expected number of cases of Down syndrome	probability of result occurring by chance
families at risk of NTD (Israel)	1492	11	1.87	p = 0.00001
group of families	number of pregnancies	number of cases of NTD	expected number of cases of NTD	probability of result occurring by chance
families at risk of Down syndrome (Ukraine)	1847	7	1.37	p = 0.001

Source: <https://www.ncbi.nlm.nih.gov/pubmed/12711468>

(b) With reference to Table 2.1, justify the claim that there is a link between Down syndrome and NTD in these families. [3]

- Families at risk of NTD in Israel has a higher number of Down's syndrome cases of 11 than expected number of 1.87.
- Families at risk of Down's syndrome in Ukraine has a higher number of NTD cases of 7 than expected number of 1.37.
- The observed numbers of Down's syndrome and NTD in both families in Israel and Ukraine are significantly different from the expected number where both are less than p = 0.05.

(c) NTD is due to recessive mutation on a gene found on chromosome 9.

Complete Table 2.2 to show the results of the calculations.

[2]

Table 2.2

parent 1	parent 2	percentage probability of having a child with NTD	percentage probability of having an unaffected child (not carrier)	percentage probability of having a child who is a carrier
unaffected	carrier	0	50	50
carrier	carrier	25.....	25.....	50.....
unaffected	has NTD	0.....	0.....	...100...
carrier	has NTD	50	0	50

(d) Scientists investigated the use of induced pluripotent stem cells (iPS cells) to treat NTD in mice. The procedures are as follows:

- The skin of mice with NTD were collected within the first few hours of birth
- The skin-derived fibroblast cultures were established
- Plasmids containing four transcription factors to reprogramme skin cells to form iPS cells were introduced into the fibroblast cultures.
- The scientist then stimulated the *in vitro* differentiation of iPS cells into normal neural cells.
- The normal neural cells were transplanted back to mice.

Explain how transcription factors can reprogramme cells to form iPS cells.

[2]

- transcription factor/ activator binds to a control element/ enhancer of a gene
- .. upregulate the expression of gene that codes for proteins to increase the pluripotency of the fully differentiated stem cells.

OR

- transcription factor/ silencer binds to a control element/ repressor of a gene
- ..downregulate the expression of gene that codes for proteins to decrease/ inhibit the differentiation of stem cells.

- (e) The iPS cells were first discovered in 2006 and have been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine.

Recent research shows that abnormal reprogramming can occur during the induction of human induced pluripotent stem cells (HiPS cells) and tumours could be generated.

- (i) Compare the embryonic stem cell and HiPS cell. [2]

Difference

- Embryonic stem cell is obtained from inner cell mass of blastocyst / embryos whereas HiPS cells are obtained from differentiated somatic cell

Similarities

- Both are pluripotent / have the potential to become any cell type in the adult body but not those of the extra-embryonic membranes
- Both have self-renewing capabilities / can divide continually
- Both are unspecialised/undifferentiated

- (ii) Comment on the ethical issues involved in the use of HiPS cells. [2]

- Overcome ethical issue of using embryonic stem cells for treatment as it does not involve killing of embryo.
- No tissue rejection involved as the HiPS cells are derived from the patient
- [idea of] Treatment to benefit patients/ alleviate the suffering of many people
- [idea of] Safety concerns on the use of HiPS cells.
- Potential to develop into a human embryo/ to clone human being/produce germ cells

[Total: 13]

QUESTION 3

Immediately following an infection with human immunodeficiency virus (HIV), there is a period of high levels of HIV in the blood (viremia). HIV is an intracellular pathogen and would stimulate a strong CD8 cytotoxic T cell response. Antibodies against HIV glycoproteins and nucleocapsid proteins are also produced. The immune response against HIV results in a reduction in viremia but does not eliminate the virus completely.

- (a) It is known that HIV infects CD4 T helper cells. Suggest other features about HIV that make it difficult for the immune response to eliminate it. [3]

- The HIV double-stranded DNA can integrate into the host genome to form provirus and lie dormant within the host cell, thus, evades the detection by immune response.
- Reverse transcription is prone to errors and reverse transcriptase does not proofread.
- ...HIV has very high mutation rate / antigenic drift in the gene coding for viral glycoprotein, hence antibodies can no longer complementarily bind to the mutated viral glycoprotein.
- HIV has very high replication rate which produces more viral particles than it can be eliminated.

Fig. 3.1 shows a test that has been developed to find out if a person has antibodies to the HIV antigen.

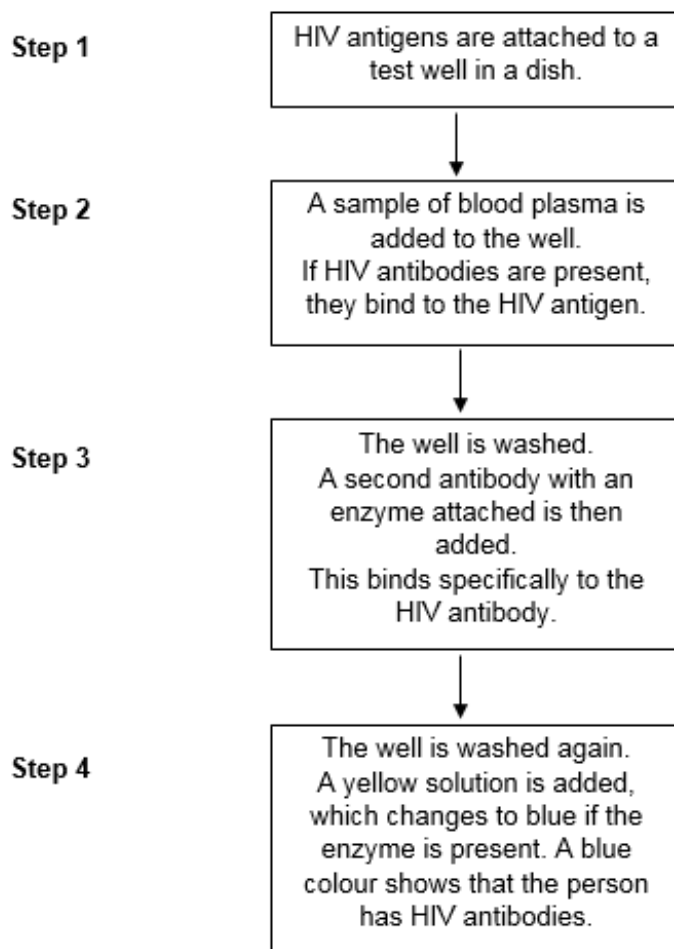


Fig. 3.1

(b) This test only detects the presence of HIV antibodies. Suggest why it cannot be used to find out if a person has AIDS. [1]

- To diagnose AIDS, need to look for AIDS-related symptoms (ORA)
- ..look at number of helper T cells

(c) Explain why the solution will remain yellow if a person is **not** infected with HIV. [2]

- HIV antibody is not present
- ..second antibody will not bind to the HIV antibody/ enzyme is not present

(d) A mother who was infected with HIV gave birth to a baby. The baby was tested positive using this test. Suggest why this does not prove that the baby is infected with HIV. [2]

- Children receive HIV antibodies from their mothers
- ..so solution will always turn blue/ will always test positive
- Children have not produced antibodies yet so test may be negative if they did not receive the HIV antibodies from the mother

[Total: 8]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 4

- (a)** Evolutionary studies suggest that some membrane-bound organelles in plant cells have evolved from a symbiotic relationship between eukaryotic and prokaryotic cells. Molecular phylogenetic techniques revealed similarities between the structures of the organelles and that of the prokaryotic cells.

Describe how structures involved in photosynthesis provide evidence for endosymbiosis and outline how molecular techniques such as gel electrophoresis can be used to show this phylogenetic relationship between prokaryotes and eukaryotes. [10]

- (b)** Climate change has been attributed to a rise in carbon dioxide levels, which may affect organisms directly or indirectly.

Explain how an increase in carbon dioxide levels may affect plants directly and indirectly. [15]

[Total: 25]

QUESTION 5

- (a)** All cells, from bacteria to humans, express their genetic information via the central dogma of molecular biology which depicts the flow of genetic information from DNA to protein.

Outline the processes involved in the central dogma of molecular biology, and explain, using specific examples, why the central dogma may not always hold true. [10]

- (b)** The expression of genes gives rise to functional proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of proteins in the control of responses in organisms. [15]

[Total: 25]

QUESTION 4

- (a) Evolutionary studies suggest that some membrane-bound organelles in plant cells have evolved from a symbiotic relationship between eukaryotic and prokaryotic cells. Molecular phylogenetic techniques revealed similarities between the structures of the organelles and that of the prokaryotic cells.

Describe how structures involved in photosynthesis provide evidence for endosymbiosis and outline how molecular techniques such as gel electrophoresis can be used to show this phylogenetic relationship between prokaryotes and eukaryotes. [10]

Structures involved in photosynthesis as evidence for the endosymbiosis theory (at least 2)

1. Presence of circular DNA in chloroplast which is also found in prokaryotic cell
2. Presence of 70S ribosomes in chloroplast which is also present in the prokaryotic cell
3. Presence of double membrane in chloroplast which suggests that the ancestral prokaryotic cell was engulfed by a eukaryotic cell.
4. Size of chloroplast is similar to the prokaryotic cell

Gel electrophoresis to show the phylogenetic relationship

5. **[Compulsory]** Samples of DNA from several related species of prokaryotes and eukaryotes are obtained and the homologous DNA region of each species are isolated (by PCR)
6. Mix the DNA samples with loading dye/ glycerol and bromophenol blue.
7. Load the DNA samples into the wells of the agarose gel near the cathode/ negative end.
8. A DNA ladder is added to one well to allow for the comparison of the size of the fragments.
9. When the electricity is turned on, these negatively charged DNA molecules migrate towards the positive end of the field/anode.
10. The DNA molecules are separated according to their molecular size
11. The smaller molecules are able to move through the agarose gel faster than the larger one, so they will travel further in the gel than the larger molecules (and vice versa).
12. Visualise the DNA by soaking the agarose gel in ethidium bromide / fluorescent DNA binding dye which binds to the DNA strands in the gel and view under UV light
13. Compare the positions of the bands between the samples and estimate the size of fragments in the band against the DNA ladder
14. The more similar the band pattern of the rRNA gene/ homologous DNA of the eukaryotes and prokaryotes, the more closely related they are.
15. Submerged gel in buffer to maintain pH / allow electricity to pass through.

QWC [1]

Correct description of at least two structures as evidences on endosymbiosis theory, clear outline of gel electrophoresis technique in correct sequence to determine the phylogenetic relationship.

- (b) Climate change has been attributed to a rise in carbon dioxide levels, which may affect organisms directly or indirectly.

Explain how an increase in carbon dioxide levels may affect plants directly and indirectly. [15]

Overview of the answers

Part I: Direct effects of increasing carbon dioxide concentration on plants [max 5]

- Increase in carbon dioxide increases the rate of light-independent reaction of photosynthesis.
- Increases the yield of crop

Part II: Indirect effects of increasing carbon dioxide concentration on plants [max 9]

- Role of carbon dioxide in producing global warming
- Global warming affects:
 - distribution and adaptation of plants
 - global food supply e.g. crop yields

Part I. Direct effects of increasing carbon dioxide concentration on plants [max 5]

1. Increase in carbon dioxide increases the rate of light-independent reaction of photosynthesis.
2. *[CO₂ fixation]* One molecule of CO₂ combines with a 5C ribulose biphosphate (RuBP) molecule to give a 6-carbon intermediate catalysed by enzyme RUBISCO
3. This 6C intermediate is **unstable**, hence it will break down to give two molecules of 3C **glycerate-3-phosphate (GP) / 3-phosphoglycerate (PGA)**.
4. *[PGA reduction]* Each molecule of **PGA/GP** receives 1 additional phosphate group from an **ATP** molecule to form **glycerate-1,3-bisphosphate / 1,3-bisphosphoglycerate**.
5. Each molecule of **glycerate-1,3-bisphosphate** is **reduced by NADPH** from the light-dependent reactions. **Glycerate-1,3-bisphosphate** also **loses a phosphate group** to form **glyceraldehyde-3-phosphate (GALP) / triose phosphate (TP)**.
6. A small proportion of GALP/TP produced is used to **synthesize carbohydrates**.
7. *[RuBP regeneration]* The remaining GALP/TP is used to **regenerate RuBP** using ATP from light dependent reaction
8. *[idea of]* Increases the yield of crop

Part II. Indirect effects of increasing carbon dioxide concentration on plants [max 9]

[Role of carbon dioxide in producing global warming]

9. CO₂ gas traps heat in the atmosphere resulting in global warming.

[Effects of global warming on distribution and adaptation of plants]

Plant distribution

10. Habitats of some types of plants are likely to **move northward** or to **higher altitudes**.
11. Other species may **die** if conditions in their current geographic range are no longer suitable.

Plant adaptations

12. **Closing** and **reduction** in the number of **stomata** per leaf to **reduce the loss of water by transpiration** due to increase in temperatures
13. Some **leaves narrowed in width/thickness** to help to **decrease water loss by transpiration**
14. Increase in temperatures has **negative effects on plant growth and development** by disrupting the stability of various proteins, membranes, and cytoskeleton structures.
15. Alterations in **membrane lipid composition** – increase % of saturated fatty acids in membrane lipids increases lipid melting temperature, hence, prevent heat-induced increase in membrane fluidity.

[Effects of global warming on global food supply e.g. crop yields]

16. **physiological injuries of the plants** observed under **elevated temperatures** (e.g. fruit damage) leading to a **decreased plant productivity/ crop yields**.
17. Animal that feed on plants may move northward, decreases the crop yield
18. In some crops (e.g. grains), **faster growth reduces the amount of time that seeds have to grow and mature**. This can **reduce yield**.
19. **Warming/increased in temperature that exceeds a crop's optimum temperature/ causes denaturation of enzymes**, hence, **reduces crop yields**.
20. **Increased in temperature increases the growth of weeds, pests and fungi** which would affect crops previously unexposed to these species, hence, reduces the crop yield.
21. Competition for food resources/ habitat as a result of migration of plant northward
22. Global warming causes **flooding** which kills crops
23. Global warming affects the **metabolic rate of pollinators/insects**, may result in increase or decrease pollination of flowers, hence, affects crop yield.
24. Global warming may affects **flowering seasons**, which may affect development and growth of plant.
25. Global warming causes **sea level to rise**, increases **salinity of freshwater**, affects crop yield
26. Global warming results in **drought**, decreases crop yield
27. CO₂ dissolves in seawater to cause **ocean acidification**, causes plants to die./ dissolve in freshwater to cause **acid rain**, causes plants to die.

QWC [1]

Communicate clearly how increase in CO₂ level affect plants directly and indirectly

QUESTION 5

- (a) All cells, from bacteria to humans, express their genetic information via the central dogma of molecular biology which depicts the flow of genetic information from DNA to protein.

Outline the processes involved in the central dogma of molecular biology, and explain, using specific examples, why the central dogma may not always hold true. [10]

[Transcription]

1. General transcription factors and RNA polymerase bind to the promoter
2. One of the DNA strands is used as the template to synthesize mRNA
3. Template read from 3' to 5' / mRNA synthesized from 5' to 3'
4. RNA polymerase adds ribonucleotides by complementary base pairing via hydrogen bonds
5. Catalyzes formation of phosphodiester bonds between adjacent ribonucleotides.

[Translation]

6. Mature mRNA used as a template to synthesize polypeptide
7. mRNA read from 5' to 3'
8. Ribosome peptidyl transferase catalyses peptide bond between adjacent amino acids
9. Each codon codes for one amino acid
10. Punctuated: AUG start codon, UAA/UGA/UAG stop codon

[Why Central Dogma is not always true- at least 2]

11. Flow may be reverse, from RNA to DNA
 - a. **Ref. to HIV:** (+)ssRNA reverse transcribed to ssDNA
 - b. **Ref. to telomerase:** telomerase RNA as the template for reverse transcribed to DNA (telomere)
12. Flow may be from RNA to RNA
 - a. **Ref. to influenza:** (-)ssRNA transcribed to (+)ssRNA
13. Flow may end at RNA
 - a. **Ref. to** tRNA / rRNA / snRNA / telomerase RNA

QWC [1]

Central Dogma processes communicated accurately and to include at least two specific examples

- (b) The expression of genes gives rise to functional proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of proteins in the control of responses in organisms.

[15]

Cell signalling /homeostasis

1. [Importance] Protein hormones act as ligands/chemical messengers to activate signaling pathways
2. [Importance] Receptor proteins on cell surface membrane where specific ligands bind to via complementary shape to initiate signal reception to trigger transduction pathways and cellular response
3. [Importance] G protein/relay proteins transduces signals from activated receptors to other proteins in the cell resulting in amplification cascade
4. [Example] insulin binds to tyrosine receptor kinases on the effector cells to decrease blood glucose concentration
5. [Example] glucagon binds to G-protein coupled receptors on the effector cells to raise blood glucose concentration
 - a. reference to any **enzyme** involved in cellular response e.g. protein kinase, adenylyl cyclase

Immune response

a) Innate immune response

6. [Importance] Toll-like receptors on phagocytic cells recognize and bind to a variety of microbial products eg. bacterial or fungal components
7. [Importance] Production of **antimicrobial proteins** are triggered upon pathogen recognition to **attack pathogens or impede their reproduction.**
8. [Example] **interferons/ complement proteins.**
9. [Example] **Hydrolytic enzymes** in the lysosome degrade the components of the pathogens
10. [Example] **Lysozyme** in tears, saliva, and mucous secretions **destroys cell walls** of susceptible bacteria/ involved in phagocytosis to ingest pathogen

b) Adaptive Immune response

11. [Importance] Presence of specific antigen receptor protein on B cell /T cell binds to specific antigen e.g. on the surface of pathogen via **complementary shape**
12. [Example] Major histocompatibility complex (MHC) proteins on antigen-presenting cells or non-immune cells present fragments of antigens on the surface of host cells so that T cell receptors can bind to
13. [Example] Cytokines (protein) produced from antigen-presenting cell and helper T cell stimulate the helper T cell to proliferate.

14. [Example] Helper T cells produce other cytokines to activate cytotoxic T cells to kill infected cells
15. [Example]..and stimulate antigen-activated B cell to proliferate and differentiate into memory B cells and plasma cells that secretes antibodies

Control of gene expression – in eukaryotes

16. At chromosomal DNA level by histone modification/DNA methylation using proteins/enzymes
17. At transcriptional level by specific transcription factors e.g. activators/repressors
18. At post-transcriptional level by addition of 5' methy-guanosine cap of mRNA/ polyadenylation at 3' end of mRNA/RNA splicing/ alternative RNA splicing using proteins/enzymes
19. mRNA degradation by enzymes
20. At translational level by regulation of translation initiation using proteins/enzymes
21. At post-translational level by protein modification e.g. phosphorylation / protein tagged by ubiquitin is degraded by proteasome

Control of gene expression – in prokaryotes

22. CRP binds to CRP-binding site, upregulates expression of structural genes in the operon
23. Active repressor protein binds to operator to switch off gene expression of the operon

QWC[1]

Importance of proteins communicated clearly without ambiguity and to include at least two different control of responses in organisms

😊 End of Paper 3 😊